

ZHEN HAO GAN

A*STAR National Science Scholarship (NSS-PhD, 5-yr fully-funded Ph.D. study)

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RESEARCH INTERESTS

Embodied Navigation with Skills Adaptation under different Morphologies & Kinodynamics | Embodied Physical Intelligence | 3D Task & Motion Planning in Unknown-Unstructured Environments | Autonomous Strategy Selection with Kinodynamic Feasibility | Legged & Humanoid Robots

EDUCATION

University of Michigan, Ann Arbor, Michigan, USA

Aug 2025 - Present

PhD in Robotics

- Advised by: Prof. Bernadette Bucher

Nanyang Technological University (NTU), Singapore

Aug 2017 – Jun 2021

Bachelor of Engineering (Electrical and Electronic Engineering)

- CGPA:** 4.80/5.00, Honours (Highest Distinction)
- 2x Dean's List - Top 5% of the Cohort** (2018 – 2019, 2019 – 2020)
- Advisor: Prof. Xie Lihua

Technical University of Munich (TUM), Munich, Germany

Oct 2019 – Mar 2020

Student Leader, Winter Semester Exchange (Fakultät für Elektrotechnik und Informationstechnik)

- Postgraduate Module: (1) Computational Intelligence (2) Broadband Communication Network (3) Principle in Electrotechnology (4) Computer Architecture and Network

PUBLICATIONS

- Gan, Zhen Hao**, H. P. Palanivelu, J. J. Pey, and M. Y. Yee, Chuah, “Towards active perception: Dynamic probing for safe robot locomotion using a digital double framework,” 2025, *in submission*.
- Gan, Zhen Hao**, Y. You, and M. Y. Yee, Chuah, “Navigation with tactile sensor for natural human-robot interaction,” [arXiv preprint arXiv:2211.13091](#), 2022, *in submission*.
- Y. C. Ng, Q. Wen, C. Y. Tan, **Gan, Zhen Hao**, M. Yee, *et al.*, “Tactile aware dynamic obstacle avoidance in crowded environment with deep reinforcement learning,” [arXiv preprint arXiv:2406.13434](#), 2024.

RESEARCH EXPERIENCE

Embodiment-Aware Transformer for Legged Robot, I^2R , A*STAR

Oct 2024 – Present

Robotics Research Engineer, advised by Dr. Michael Chuah

- Leading research on Embodiment-Aware transformers for legged robots, focusing Connectivity-Informed, Structure-Aware learning for enhanced Physical Adaptation and Self-Modelling, including quadrupeds and humanoids.
- Developing Zero-Shot, Fault-Tolerant locomotion policies for cross-embodiment generalization with Graph Physical Bias.
- Leveraging development of this project as cornerstone for Physical Intelligence and building towards Embodied Intelligence.

3D Terrain Reconstruction with Semantic Information, I^2R , A*STAR

Oct 2023 – Present

Robotics Research Engineer, advised by Dr. Michael Chuah

- Proposed ideas on 3D Reconstruction of Point Cloud based on semantic information of terrains to improve on traversability performance and generalize to more complicated outdoor terrains.
- Developing high speed, low computation cost 3D Terrain Reconstruction Module with Semantic Information in real-life.
- Eliminate noisy raw input and fill missing data in scenarios like tall grasses and grates to avoid rogue locomotion and smoothen behavior when compared to previous work (*achieved 80% success rate in these terrains*).
- Enable legged robots to respond correctly on the terrains with the reconstruction module.

Real-Time Dynamic Probing Framework for Legged Robots, I^2R , A*STAR

Oct 2023 – Present

Robotics Research Engineer, advised by Dr. Michael Chuah

- Investigate the feasibility of providing collapsibility and terrain information through the combination of joint-level proprioceptive information on legged robots and reasoning from real-time dynamic probing on suspicious terrains.
- Developing a framework to incorporate the real-time digital double for collapsibility estimation (via Conv1D model) using impulse reaction force through the robot's foot and greater than the robot's weight without exceeding the foot's limits.
- Integrate Path and Motion Planner based on the output from this framework to avoid dangerous terrain and achieve safe legged robot navigation in unstructured-uneven terrains.

Navigation with Tactile Sensor for Natural Human-Robot Interaction, I^2R , A*STAR Dec 2020 – Jun 2021

Robotics Research Engineer, advised by Dr. You Yangwei & Dr. Michael Chuah

- Investigated the use of tactile sensors in robotic navigation stack and propose new social navigation stack with tactile sensors.

- Developed social robotic navigation stack for human-crowded environments with tactile sensors, RGB camera & LiDAR.
- Completed robot with social navigation abilities: proximity filter compliance control, active social interactions, and passive social interactions using force responses, dynamic cost assignment and semantic segmentation.
- Built semantic layers for robots to behave with socially acceptable reactions by both semantic & force information.

Vision-based Robot Navigation via Deep Reinforcement Learning, NTU

Jul 2020 – Jun 2021

Final Year Project, advised by Prof. Xie Lihua

- Developed visual perception module for pedestrian identifier using YOLOv4.
- Built and tested *DRL-Based Autonomous Navigation* through Python to fuse camera data and GA3C- CADRL for AGVs navigation in pedestrian-rich environment using ROS-Gazebo Simulation.
- Created and tested *Pedestrian Interaction Module* with Collision Avoidance System for DRL navigation using PyTorch.
- Achieved >80% success rate when navigating around 10 human obstacles in a small area.

PROFESSIONAL EXPERIENCE

Robotics Research Engineer, I²R, A*STAR

Sep 2021 – Present

supervised by Dr. Michael Chuah & Dr. Albertus Hendrawan Adiwahono

- **Project Rover-X:**
 - Developed and implemented GPS Goal-driven Exploration Module to allow legged robot exploration in unmapped areas.
 - Developed Traversability Analysis & Interpreter modules for autonomous gait adaptation on treacherous terrains.
 - Successfully delivered D6 fully autonomous patrolling and people following demo to HTX & SCDF with replacement of human operators in high-risk HazMat operations. [Featured at The Straits Times \(Singapore's largest news outlet\).](#)

HONOURS & AWARDS

- **2024:** A*STAR National Science Scholarship (**NSS-PhD, 5-yr funding for Ph.D. study**), Singapore
- **2019:** NTU EEE Partial Financial Award for GEM Explorer
- **2019:** Dean List's Academic Year 2018/2019 (Top 5% of the Cohort)
- **2018:** Dean List's Academic Year 2017/2018 (Top 5% of the Cohort)
- **2017:** GCE A-Level High Achiever Award (4A*)
- **2017:** Silver Medal – Kangaroo Maths Competition (KMC)
- **2016:** Distinction – Australian Maths Competition (AMC)
- **2016:** Sunway College Special Scholarship

INTELLECTUAL PROPERTY / PATENTS / RESEARCH GRANTS

- "Towards Physical Intelligence: Embodiment-Aware Transformer Model for Legged Robots", *2024 A*STAR Career Development Fund (CDF), up to SGD \$250k, Co-I*
- "Embodied Intelligence: Revolutionizing Robotics via Common-Sense Grounding + Skills Learning", *A*STAR Embodied Intelligence Task Force, up to SGD \$100 million (lead by Dr. Michael Chuah, Prof. Daniela Rus, Prof. Sangbae Kim, Prof. Shi Fan, Dr. Cheston Tan, Dr. Yau Wei Yun)*
- Palanivelu Hari Prasanth, **Gan Zhen Hao**, Shervina Lim Qi Wen, Ng Yung Chuen, Michael Chuah Meng Yee, Albertus Hendrawan Adiwahono, "Perception Locomotion for Quadrupedal Robots using non-linear Model Predictive Control", *SWIP-PLQR-2023-29, A*STAR*
- Palanivelu Hari Prasanth, Garen Haddeler, Ng Yung Chuen, **Gan Zhen Hao**, Shervina Lim Qi Wen, Michael Chuah Meng Yee, Albertus Hendrawan Adiwahono, "A Novel Real-time Digital Double to Estimate Collapsible Terrains for Legged Robots", *SWIP-ND2R-2023-30, A*STAR*

LEADERSHIP & CO-CURRICULAR ACTIVITIES

Studying Committee, A*STAR Scholars' Network (ASN)

Sep 2024 – Sep 2025

Organising Committee Member

- Organize various events to encourage healthy and meaningful development of A*STAR scholars (mental, career, financial).
- Build an active A*STAR scholars network and connects A*STAR scholars from all disciplines in the RIE ecosystem.
- Provide professional and social support network for current and deployed A*STAR scholars.

37th EEE Club Management Committee, NTU

Aug 2018 – Aug 2019

Honorary Treasurer

- Spearheaded an internal reformation of EEE Club in finance department.
- Established a more structured procedure and strategized over \$18,000 budget in the finance process of EEE club.

RELEVANT COURSES

- Coursera Relevant Modules: (1) Deep Learning Specialization (2) Computer Vision (3) Reinforcement Learning Specialization (4) Self-Driving Cars Specialization (5) Machine Learning
- Udemy Relevant Modules: (1) Complete Self-Driving Car Course – Applied Deep Learning

SKILLS

Languages: English, Chinese (Mandarin), Malay, German

Programming Languages: C/C++, Python, Java, MATLAB

Softwares and Libraries: ROS & ROS2, Gazebo, Isaac Lab & Sim, PyTorch, TensorFlow, Git & Github